

organisations have also created virtual places to connect with their stakeholders, for various purposes like training, collaboration, or simply bonding! With covid physically separating people, they are turning to the metaverse to bring them together. So, from being a place for geeks to hangout, the metaverse has now become an advanced ‘fusion world’ that brings together the best of real and virtual worlds.

Once again, what’s metaverse?

Imagine the internet in three dimensions (3D)—you being able to move inside it, like you would in your physical world. That is, probably, where the metaverse is taking us. It makes use of advanced telecom infrastructure, artificial intelligence (AI), and extended reality (XR)—that is, virtual reality (VR), augmented reality (AR), mixed reality (MR), and the likes—to create online platforms that are more immersive, interactive, and interoperable than they are today.

These online platforms or virtual worlds can be accessed through apps on your mobile phone, computer, or other digital devices. You will be represented by your *avatar* in the metaverse. With VR gear, you can immerse yourself completely in the virtual world, but considering its prohibitive cost, many platforms just let users engage through touchscreens. The metaverse also has dynamic digital marketplaces, driven by the forces of demand and supply.

If you want to draw a quick, very basic parallel with the real world, money translates to digital currencies in the metaverse, while assets translate into non-fungible tokens (NFTs). Blockchain technology forms the backbone of these marketplaces, with the tokens and transactions being recorded in it. Real money can be converted to digital currencies through exchanges, and real assets can also be recorded as NFTs in the blockchain. Here is

where the blurring of boundaries really happens!

We have been covering all these aspects of the metaverse in detail in the past few issues of *Electronics For You*. Here, we walk through a couple of exciting and impactful use-cases of the metaverse.

Telemedicine: Try it on the twin first

The metaverse can bridge global divides in the availability of reliable medical care. In many remote regions, there are no skilled medical practitioners and people have to travel several kilometres to get proper medical care. Telemedicine helps not only in such situations where care is not available locally, but also in cases where a complex condition requires an expert’s intervention—and the expert is somewhere across the world.

The situation has improved remarkably post-Covid, with several medical practitioners getting used to treating simple conditions through phone and video calls. According to a Forbes report, just 43% of healthcare facilities had the ability to provide remote treatment to patients before 2020, but today that figure stands at 95%. In the coming years, the metaverse will add quality to quantity, making these online consultations more immersive and interactive. Virtual reality gives a feeling of being in a real clinic, improving the connect between a patient and a doctor. Agreed, VR gear is not yet within reach of the masses, but a small setup in each village is enough to cater to all the families there. Like internet cafes took the Web to the masses, perhaps virtual clinics will take expert treatment to them.

The metaverse can also help with psychological counselling and caregiving. Take the case of a recuperating patient, who requires daily physiotherapy sessions but is unable to travel to a healthcare facility. The therapist can extend the treat-

ment to the patient remotely, using a metaverse space, VR gear, and haptic feedback suits. One space where doctors are betting big on the metaverse is in helping patients with phobias. In the metaverse, doctors can expose patients to situations and things they fear or are allergic to, under controlled environments. They can monitor how the patients react and help them conquer their mind blocks.

Globally, doctors are experimenting with a concept called digital twinning. Research firm CBInsights defines, “A digital twin is a virtual copy of a physical entity or process, such as a patient, anatomical structure, or hospital environment. It’s modelled on data sources including electronic health records (EHRs), disease registries, wearable sensors, and more.” In telemedicine, digital twins would help doctors diagnose the problem better and provide a better prognosis. The doctors can trial the course of treatment on a digital twin before administering to the real patient. Powered by cloud data and AI, it also enables doctors to quickly compare the patients with similar cases, to treat them better. It can also potentially explain some grey areas like why certain patients respond better to a treatment than others.

Apart from profiling the patient, digital or virtual twinning technology can also be used to emulate the processes in the healthcare organisation, to make it more efficient. Companies like IBM and Dassault Systèmes are into digital twin tech. According to a report by Accenture, 66% of healthcare executives expect their organisation’s investment in intelligent digital twins to increase over the next three years. Research by MarketsAndMarkets indicates that the digital twin market is expected to grow to \$35.8 billion by 2025, healthcare and defence being amongst the top adopters.

The metaverse, with blockchain technology as its backbone, can also